Data Analysis for Community Impact:

Indigenous Data Sovereignty, Grant Applications and Reporting, and Community Decision-Making

Agenda

- Data Ethics and Sovereignty
- Introduction to Data Analysis
- Data Analytics Demo
- Using Data to Support Community Funding and Reporting

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Ethical Data Handling

Ethical Collection, Usage, and Dissemination

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Ethical Data Handling Considerations

- Collect data with a clear purpose
- Be transparent about how data will be used
- Ensure participation is voluntary and informed
- Collect only what is necessary
- Protect privacy and confidentiality
- PIPEDA (Personal Information Protection and Electronic Documents Act) applies to how organizations collect, use, and share personal information
  - Requires consent, limited use, and protection of data

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Indigenous Data Sovereignty

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Indigenous Data Sovereignty

What is it?

- The right of Indigenous peoples to own, control, access, and manage their data

Why it Matters

- Addresses historical misuse and extraction of Indigenous data, and instead ensures community benefit
- Supports self-determination and community governance. Builds trust, transparency, and accountability

What This Means in Practice

- Be clear and transparent about why data is being collected and seek informed and voluntary participation
- Use data in ways that benefit the community
- Avoid sharing or publishing data without permission and be mindful of privacy, especially in small communities

Data Analysis

A Brief Introduction

What is data analysis?

Data analysis is the process of **collecting, organizing, and examining data** to identify patterns, answer questions, and support decision-making.

- Data analysis works best with **structured data** (e.g., multiple choice, categories, numerical values)
- Structured data makes it easier to:
  - summarize results
  - compare groups
  - identify trends
- **Open-ended responses** (e.g., suggestions, comments) are still valuable, but are more difficult to analyze at scale

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Tools and Resources for Beginner Data Analysis

- Python: For analyzing data
  - Use an IDE such as Visual Studio Code or an online complier such as replit
  - Free Python learning resource
- PowerBI: For visualizing data
  - Free Power BI learning resource
- Excel: For data handling and visualization
  - Free Excel learning resource
- SQL: For database handling
  - Free SQL learning resource

- Other useful tools include SAS, Tableau, R, Machine Learning, and more.

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Data Analysis Process Overview

1. Define the Question: Identify the question you'd like to solve.
2. Collect the Data: Gather the raw data through surveys or records.
3. Clean the Data: Clean the data to remove errors or outliers.
4. Data Analysis: Start with exploratory data analysis (EDA) by calculating summary statistics (e.g., mean, median, standard deviation). Use descriptive statistics to summarize the data, inferential techniques (e.g., correlations, comparisons, regression) to draw conclusions, and, for more advanced analysis, use machine learning models such as clustering, classification, or forecasting to group data, make predictions, or identify patterns.
5. Visualization & Insights: Use charts, graphs, and reports to summarize the data to aid in interpretation and decision-making.

Data Analysis Demo

From Cleaning to Actionable Insights

The Dataset

Microsoft Excel
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This workshop uses a fictional dataset representing a powwow event with 950 attendees and 150 voluntary survey responses. The dataset was intentionally designed to reflect realistic patterns in attendance, engagement, and community impact. All data included is fully anonymized, meaning:

•No personally identifiable information (PII) is collected or used

•No real community data is disclosed

In this fictional example, the data is collected voluntarily and used by an indigenous-run organization. This approach aligns with key principles of data privacy and Indigenous Data Sovereignty, including:

•Respect for community ownership and control of data

•Avoidance of extracting or misusing real community information

•Ethical use of data for learning, capacity-building, and decision-making

Importing python libraries and the dataset

Chrome HTML Document

```python
import pandas as pd
import numpy as np
```

```python
df = pd.read_csv("powwow_survey_data.csv")
df.head()
```

	RespondentID	IndigenousStatus	Distance_km	FirstTimeAttendee	Role	AgeGroup	MoneySpent	SatisfactionScore	DiscoveryMethod
0	1	Status	4	No	Spectator	Youth	0	3	Social Media
1	2	Non-Status	30	No	Spectator	Adult	64	4	Word of Mouth
2	3	Non-Status	26	Yes	Spectator	Adult	25	5	Website
3	4	Non-Status	8	No	Spectator	Adult	0	5	Social Media
4	5	Non-Indigenous	23	No	Spectator	Youth	0	5	Social Media

- Import any useful python libraries
- Load and preview the dataset

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Exploring the Data

```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 152 entries, 0 to 151
Data columns (total 14 columns):
 #   Column             Non-Null Count  Dtype
---  ------             --------------  -----
 0   RespondentID       152 non-null    int64
 1   IndigenousStatus   152 non-null    object
 2   Distance_km        152 non-null    int64
 3   FirstTimeAttendee  152 non-null    object
 4   Role               152 non-null    object
 5   AgeGroup           152 non-null    object
 6   MoneySpent         152 non-null    int64
 7   SatisfactionScore  152 non-null    int64
 8   DiscoveryMethod    152 non-null    object
 9   WouldAttendAgain   152 non-null    object
 10  LearnedSomething   152 non-null    object
 11  FeltConnected      152 non-null    object
 12  WouldRecommend     152 non-null    object
```

- Review the structure of the dataset
- Identify key variables (age, distance, spending, etc.)
- Look for patterns and potential issues
- Ask: What questions can this data help answer?

Exploring the Data

```
df.describe()
```

	RespondentID	Distance_km	MoneySpent	SatisfactionScore
count	152.000000	152.000000	152.000000	152.000000
mean	76.500000	53.098684	39.934211	4.453947
std	44.022721	72.999842	38.089068	0.787643
min	1.000000	1.000000	0.000000	3.000000
25%	38.750000	12.000000	0.000000	4.000000
50%	76.500000	24.500000	32.000000	4.000000
75%	114.250000	65.500000	68.000000	5.000000
max	152.000000	443.000000	119.000000	10.000000

- It is important to review the data at a high-level to check for any errors or inconsistencies, as well as to view an early trends.
- Here, we see that the maximum value entered for Satisfaction Score was 10. However, the given range was 1-5 so this needs to be addressed before further analysis is completed.

Cleaning the Data

- Remove or correct invalid values
- Cap satisfaction scores at valid range (1–5)
- Remove extreme outliers (e.g., incorrect spending values)
- Ensure consistency across categories

```
df['SatisfactionScore'] = df['SatisfactionScore'].clip(upper=5)
```

We cap scores at 5 since that's the maximum valid value

```
df = df[df['MoneySpent'] <= 500]
```

We remove unrealistic values for amount of money spent, in the event of someone accidentally entering 3500 instead of 35.00, for example.

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High-level Insights

```
# Total money spent
df['MoneySpent'].sum()
```

```
np.int64(6070)
```

The total amount of money spent at vendor booths by individuals who completed the survey is $6070.

```
# % who spent money
(df['MoneySpent'] > 0).mean()
```

```
np.float64(0.6381578947368421)
```

Approximately 64% of survey respondees spent money at vendor booths.

A majority of attendees supported vendors, contributing to the local economy.

- What percentage of attendees spent money?
- How much money was spent at vendor booths?

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Analyzing the Financial Data

- Differences in spending across groups
- Contribution to local Indigenous vendors
- Identifying high-impact attendee groups

```
df.groupby('IndigenousStatus')['MoneySpent'].apply(lambda x: (x > 0).mean())
```

```
IndigenousStatus
Non-Indigenous    0.230769
Non-Status        0.632911
Status            0.733333
Name: MoneySpent, dtype: float64
```

73% of status attendees spent money at vendor booths, compared to only 23% of non-indigenous attendees.

```
# Average spend
df[df['MoneySpent'] > 0]['MoneySpent'].mean()
```

```
np.float64(62.577319587628864)
```

Of those who spent money, an average of $62.58 was spent per person.

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Analyzing Attendee Demographics

```python
df['IndigenousStatus'].value_counts(normalize=True)
```

```
IndigenousStatus
Non-Status        0.519737
Status            0.394737
Non-Indigenous    0.085526
Name: proportion, dtype: float64
```

```python
df['Local'] = df['Distance_km'] < 30

df.groupby('Local')['FirstTimeAttendee'].value_counts(normalize=True)
```

```
Local  FirstTimeAttendee
False  Yes                 0.666667
       No                  0.333333
True   No                  0.640449
       Yes                 0.359551
Name: proportion, dtype: float64
```

64% of local attendees have attended the powwow in previous years, compared to only 33% of attendees who travelled 30km or more.

- Demographic group distribution
- First-time vs repeat attendees
- Local vs non-local attendance

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Spending Insights

- How attendees heard about the event
- Differences by age group
- Identifying effective outreach channels
- Informing future marketing strategies

```
df.groupby(['AgeGroup', 'DiscoveryMethod']).size().unstack().fillna(0)
```

DiscoveryMethod	Poster	Radio	Social Media	Website	Word of Mouth
AgeGroup					
Adult	11	12	21	11	15
Child	2	2	2	4	2
Senior	4	10	2	1	11
Youth	1	2	24	2	13

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Overall Impact of the Event

```
df['FeltConnected'].value_counts(normalize=True)
```

```
FeltConnected
Yes    0.842105
No     0.157895
Name: proportion, dtype: float64
```

84% of attendees felt connected to their culture by attending the powwow.

```
df.groupby('IndigenousStatus')['LearnedSomething'].value_counts(normalize=True)
```

```
IndigenousStatus  LearnedSomething
Non-Indigenous    Yes                0.846154
                  No                 0.153846
Non-Status        No                 0.582278
                  Yes                0.417722
Status            Yes                0.533333
                  No                 0.466667
Name: proportion, dtype: float64
```

Non-Indigenous attendees reported high levels of cultural learning. Approximately 85% of non-indigenous attendees learned something about indigenous culture, history, or protocols.

- Satisfaction levels across attendees
- % who would attend again
- Cultural learning among non-Indigenous attendees
- Sense of community connection

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Why Visualization Matters

Raw data and tables can be difficult to interpret.

Data visualization helps:

- communicate insights clearly
- support decision-making
- make data accessible to everyone

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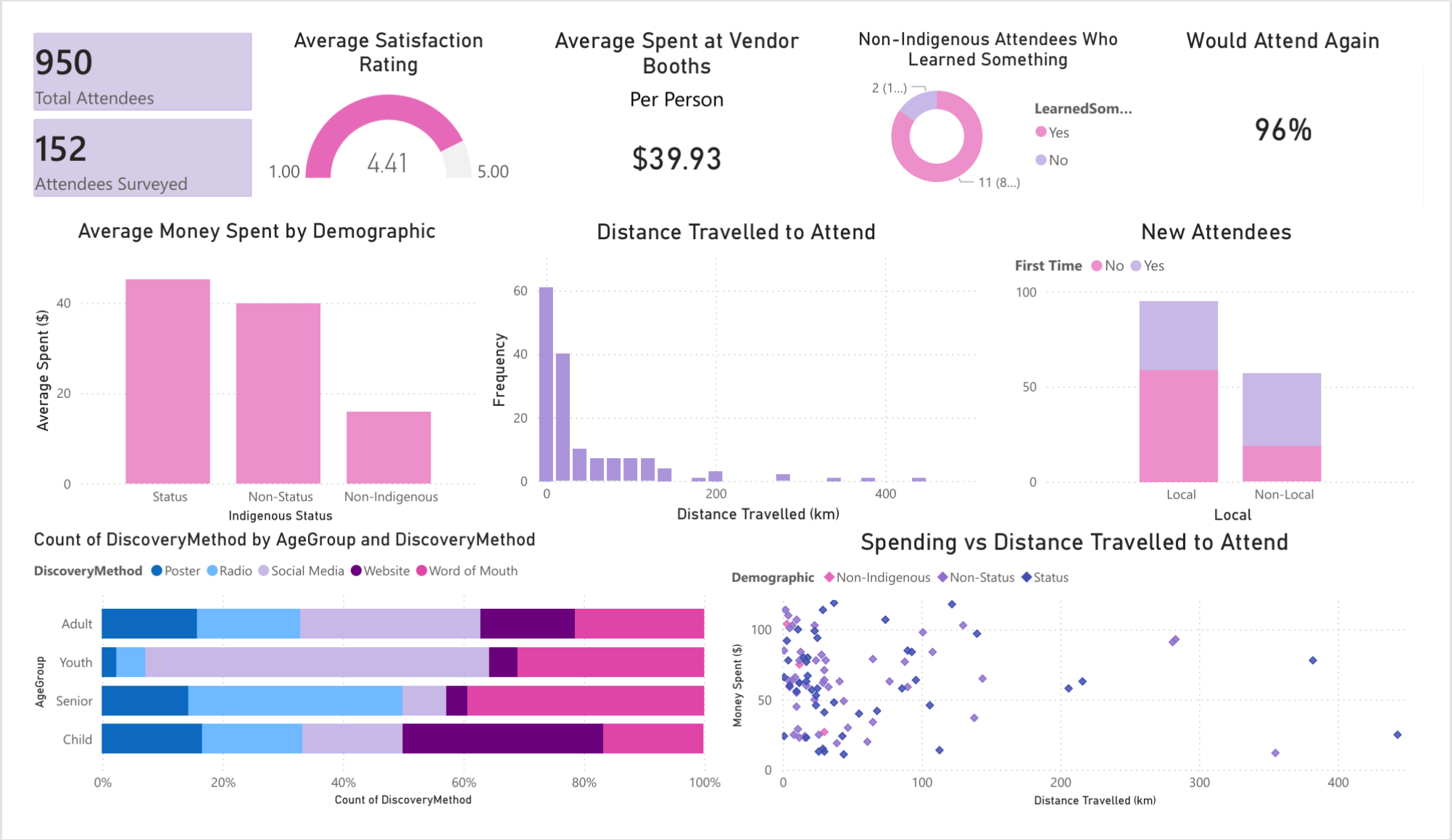
PowerBI Dashboard



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From Data to Action

Vendor Insights

- Strong vendor engagement and revenue → Attract more vendors for next year

Impact & Success

- High satisfaction and community impact → Support grant reporting & future funding

Marketing Insights

- Different groups respond to different outreach → Improve and target marketing strategies

Using Data to Support Community Initiatives

Why is it important and how can it help?

Funding Applications

- Demonstrate attendance and growth
- Show economic impact (vendor spending)
- Highlight community reach
- Provide evidence of cultural, social, or economic impact

Examples of Data You Can Include:

- "Over 950 attendees, with 150 survey responses collected"
- "65% of Indigenous attendees supported local vendors"
- "Average satisfaction score of 4.6 out of 5"
- "Attendees travelled from both local and regional communities"
- "High engagement across youth, adult, and elder groups"

Post-Event Reporting

- Summarize key metrics and outcomes
- Report to funders and stakeholders
- Identify strengths and areas for improvement
- Support future planning and decision-making

What You Can Show in Reports:

- Attendance and participation breakdowns
- Economic impact (vendor spending, local support)
- Marketing effectiveness (how people heard about the program)
- Cultural impact (learning, connection, community engagement)
- Success metrics (satisfaction, return intent)

Ideas For Useful Data to Collect:

- Participation or engagement levels (e.g., attendance, usage, involvement)
- Demographic groupings (e.g., age range, region, role)
- First-time vs repeat participation
- How participants heard about the initiative
- Satisfaction or experience ratings (e.g., 1–5 scale)
- Outcomes or impact indicators (e.g., learning, connection, skills gained)
- Optional economic indicators (e.g., spending ranges, resources use)

Final Tips & Takeaways

- Data doesn't need to be complex to be useful
- Even simple analysis can support strong decisions
- Visualizations help to communicate impact
- Ethical data practices are essential

Tips for Collecting & Handling Data

- Keep surveys short to encourage participation
- Clearly explain why you are collecting the data
- Collect only what you need and avoid unnecessary personal information
- Ensure participation is voluntary and based on informed consent
- Use consistent categories (e.g., age groups, roles) for easier analysis
- Avoid open-ended questions for analysis

Final Tips & Takeaways

Data Ethics and Sovereignty

- Follow Indigenous Data Sovereignty principles
- Ensure informed and ethical data collection
- Use data responsibly and with community benefit and informed consent

Data Safety & Responsibility

- Store data securely (password-protected files)
- Avoid sharing raw data publicly or with third parties, even if anonymized
- Be mindful of small communities where individuals may still be identifiable
- Follow applicable privacy guidelines (e.g., PIPEDA)

Thank you

Digital copy of the slideshow

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